Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: Multi-Step LLM Agents | *Python, Gemini, Scikit-learn, CP-SAT*

- Built an LLM assistant that turns scheduling requests into constraints for a solver and negotiates conflicts with people.
- A simple classifier handles routine requests quickly, and the LLM is used only for the harder ones.
- Built hybrid keyword and embedding search over policy documents, reaching **93%** recall.
- Built an evaluation suite with baselines; **no attack succeeded** in 50 prompt-injection tests.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Tested whether the models were reasoning or memorising answers before reporting results.

Football Video Analytics with YOLO | *Python, YOLOv8, OpenCV, Streamlit*

- Built a video analysis pipeline that tracks players, referees and the ball with a fine-tuned YOLOv8 model.
- Grouped players into teams by jersey colour and mapped the camera view to the pitch to measure speed and distance.
- Streamed frames in batches to cut memory use from **5.3 GB to 1 GB**, and built a Streamlit app to explore the results.

Distributed Data Pipelines: MapReduce & gRPC Streaming | *Hadoop, C++, Python, gRPC*

- Rewrote a 1M-record weather analytics job as a single Hadoop MapReduce stage.
- Built a real-time streaming pipeline with gRPC that handles about **100K records per second** with no data loss.

Technical Skills

Languages: Python, C/C++, SQL, Bash.
Machine Learning: PyTorch, Hugging Face, Scikit-learn, LoRA, RAG, YOLOv8, OpenCV.
Data: NumPy, Pandas, Hadoop, gRPC.
Tools: FastAPI, Streamlit, Linux, Git, Docker.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.